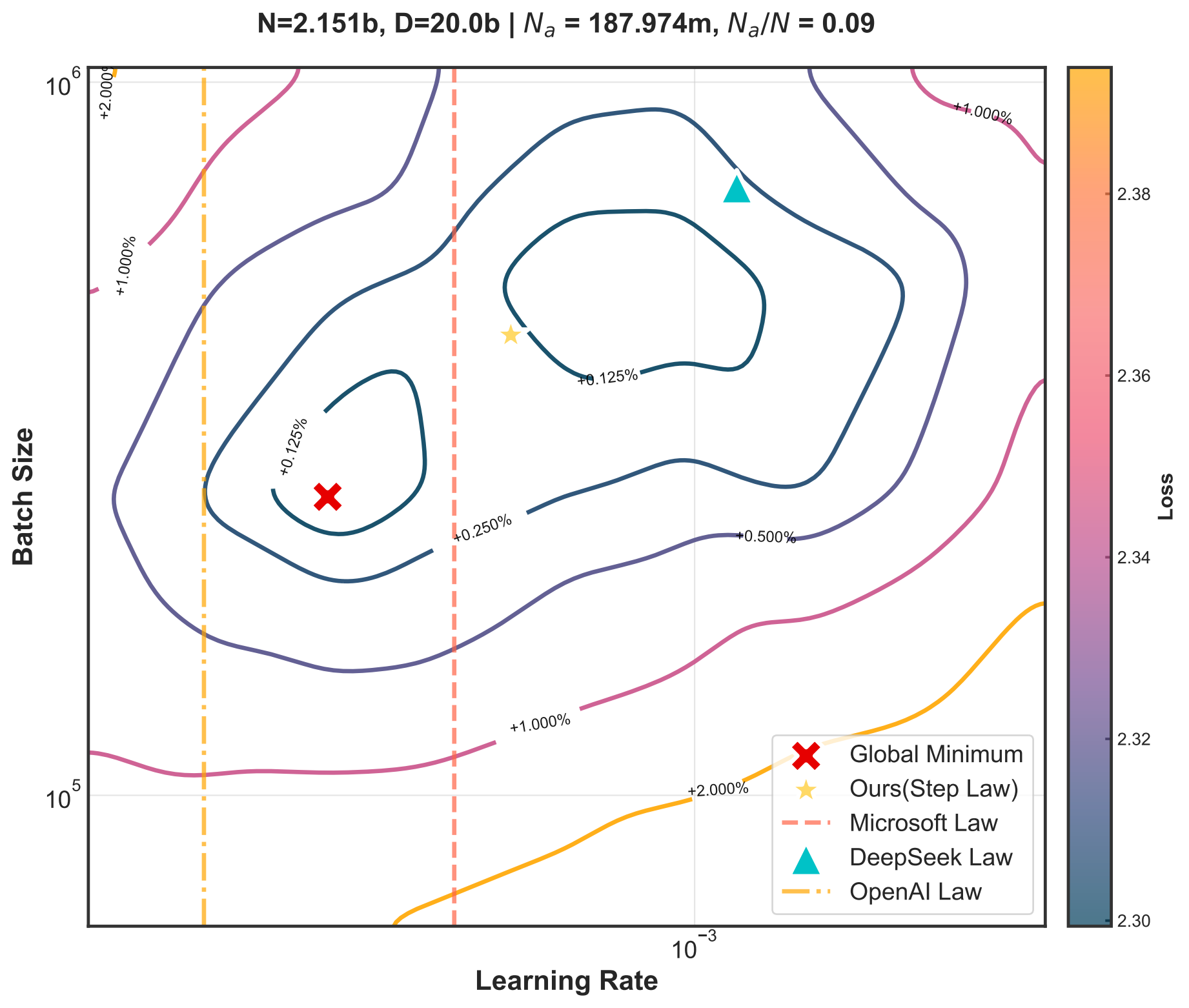
N=2.151b, D=20.0b | N_a = 187.974m, N_a/N = 0.09
Batch Size
Learning Rate
Loss
10^6
10^5
10^{-3}
2.38
2.36
2.34
2.32
2.30
+2.000%
+1.000%
+0.125%
+0.250%
+0.500%
+1.000%
+2.000%
+1.000%
+0.125%
Global Minimum
Ours(Step Law)
Microsoft Law
DeepSeek Law
OpenAI Law